



CSE 251

Electronic Devices and Circuits

Lecture 10

Course instructor:

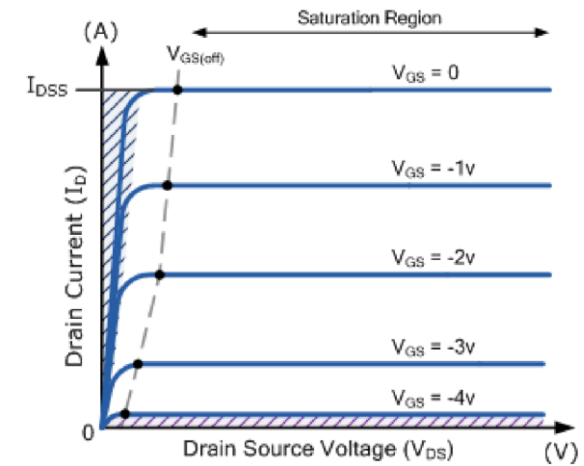
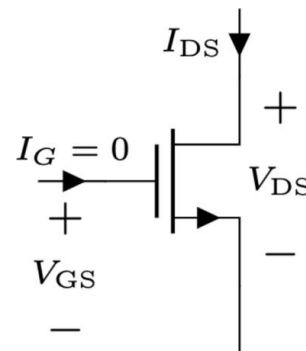
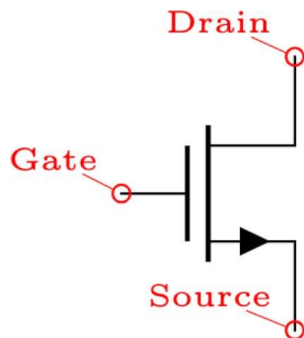
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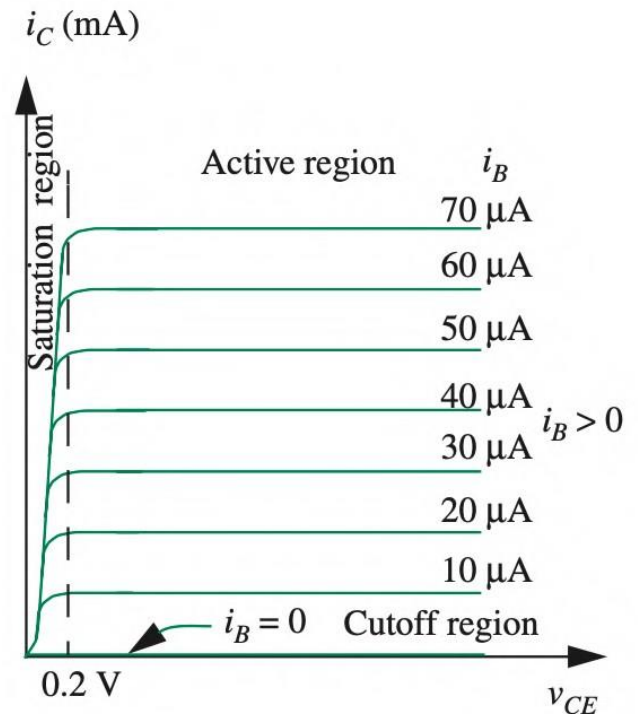
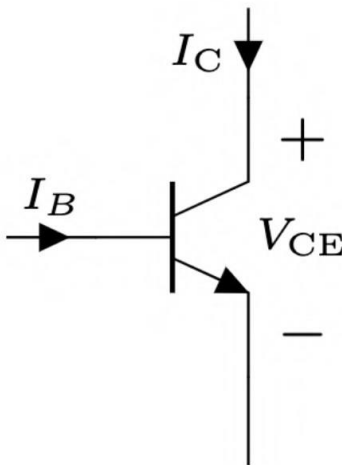
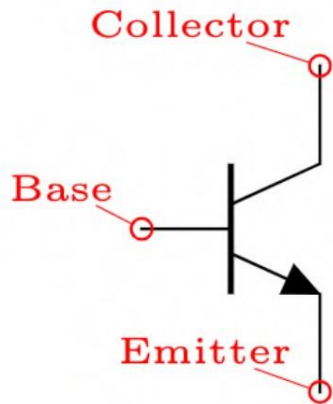
Transistors as Digital Switch

- Transistors are 3 terminal non-linear devices, can be used as switch
- 2 types – **Voltage Controlled**, **Current Controlled**
- **M**etal **O**xide **S**emiconductor **F**ield **E**ffect **T**ransistor (**MOSFET**) are **voltage controlled**
- Control, $C = V_{GS}$. The IV characteristics (I_{DS} vs V_{DS}) depends on V_{GS}
- Actual dependency is complex.
- Will start with a simple (but approximate) one – **S-Model** (Switch Model)



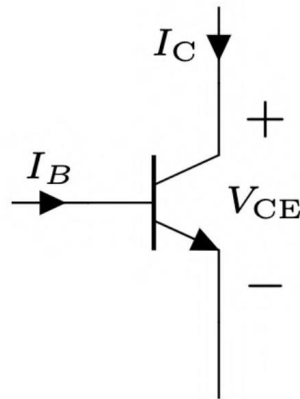
Bipolar Junction Transistor

- **Current-controlled** transistor, 3 terminals – Base, Emitter, Collector
- IV between C and E (I_C vs V_{CE}) is controlled by base current, I_B
- IV is quite like MOSFET, but there are some differences
- We can use a S-model here too, but controlled by I_B (instead of V_{GS})



BJT vs MOSFET - Differences

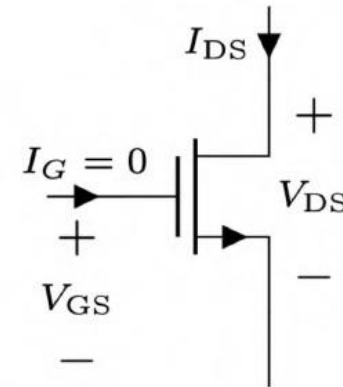
BJT



Current controlled, I_B controls (I_C vs V_{CE})

Base current, I_B , is the control. Hence $I_E \neq I_C$, rather
 $I_E = I_C + I_B$

MOSFET

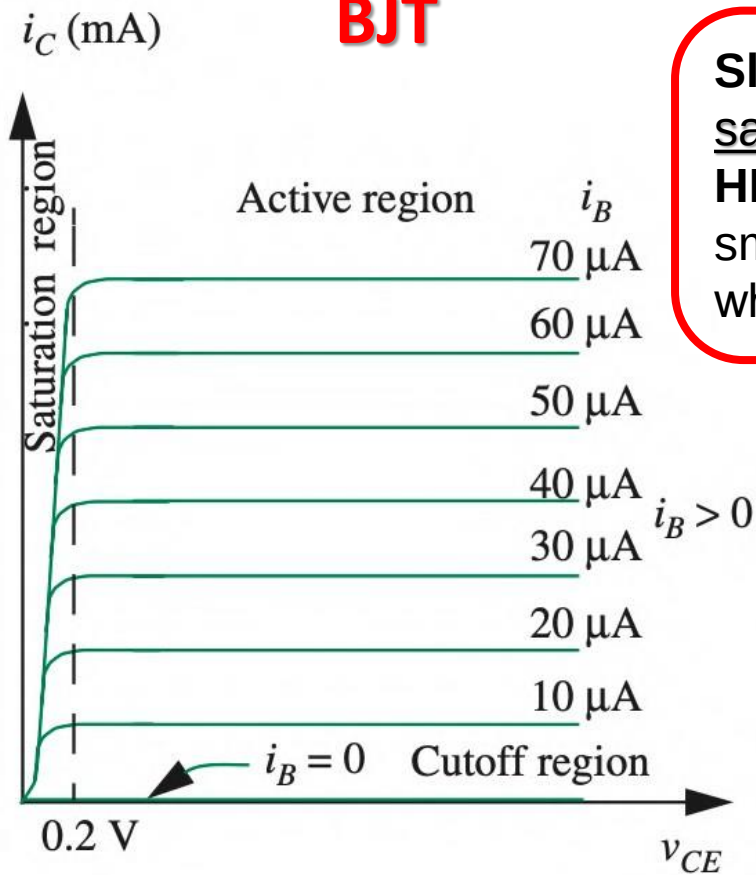


Voltage controlled, V_{GS} controls (I_{DS} vs V_{DS})

Gate current, I_G , is always 0. Hence $I_S = I_D = I_{DS}$.

BJT vs MOSFET - Differences

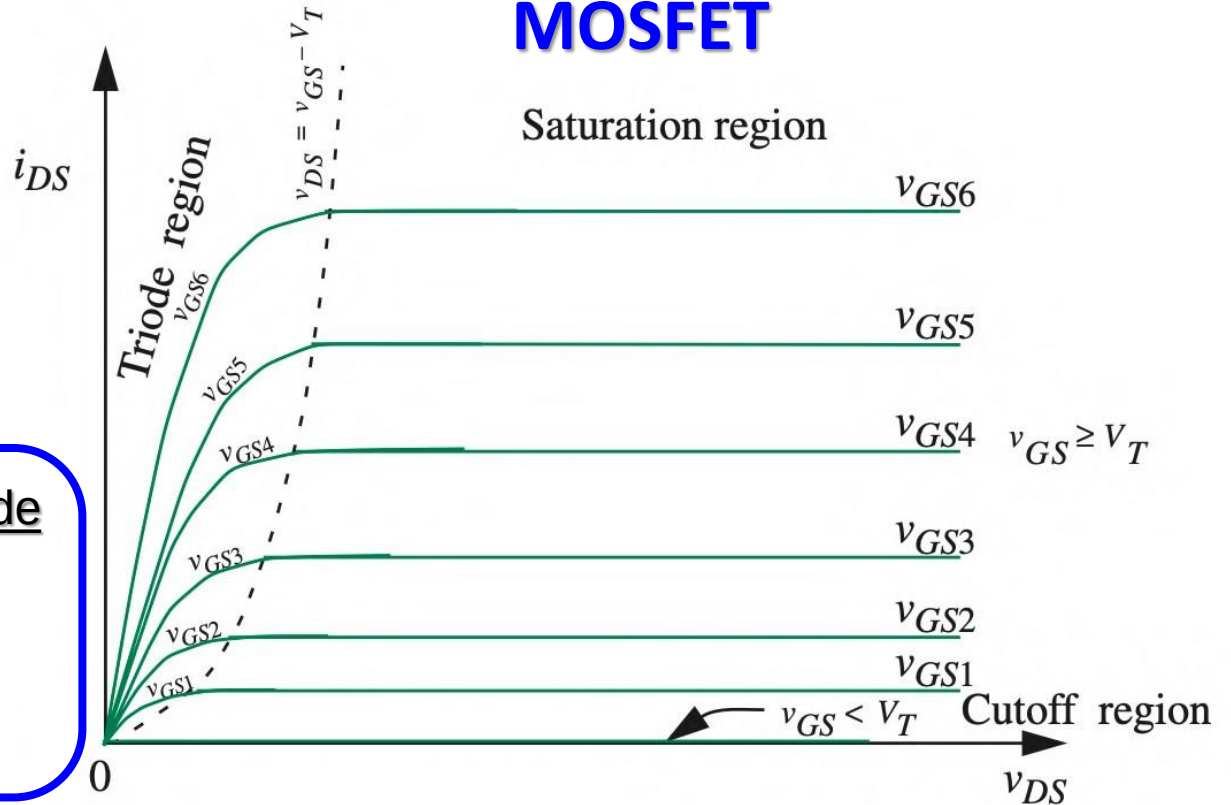
BJT



Slope in the saturation region is **HIGH**, hence very small resistance when "ON"

Slope in the Triode region is **LOW**, hence significant resistance when "ON"

MOSFET

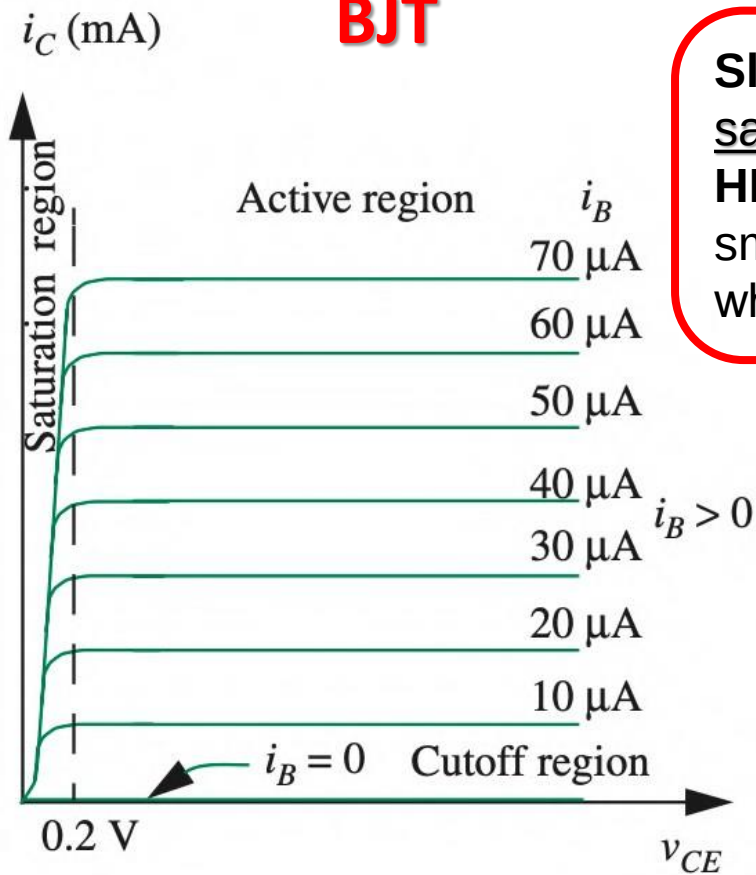


Current in **Saturation** region changes quadratically with control V_{GS} .
Hence, $I_{DS} \propto V_{GS}^2$

Current in **active** region changes linearly with control I_B .
Hence, $I_C \propto I_B$

BJT vs MOSFET - Differences

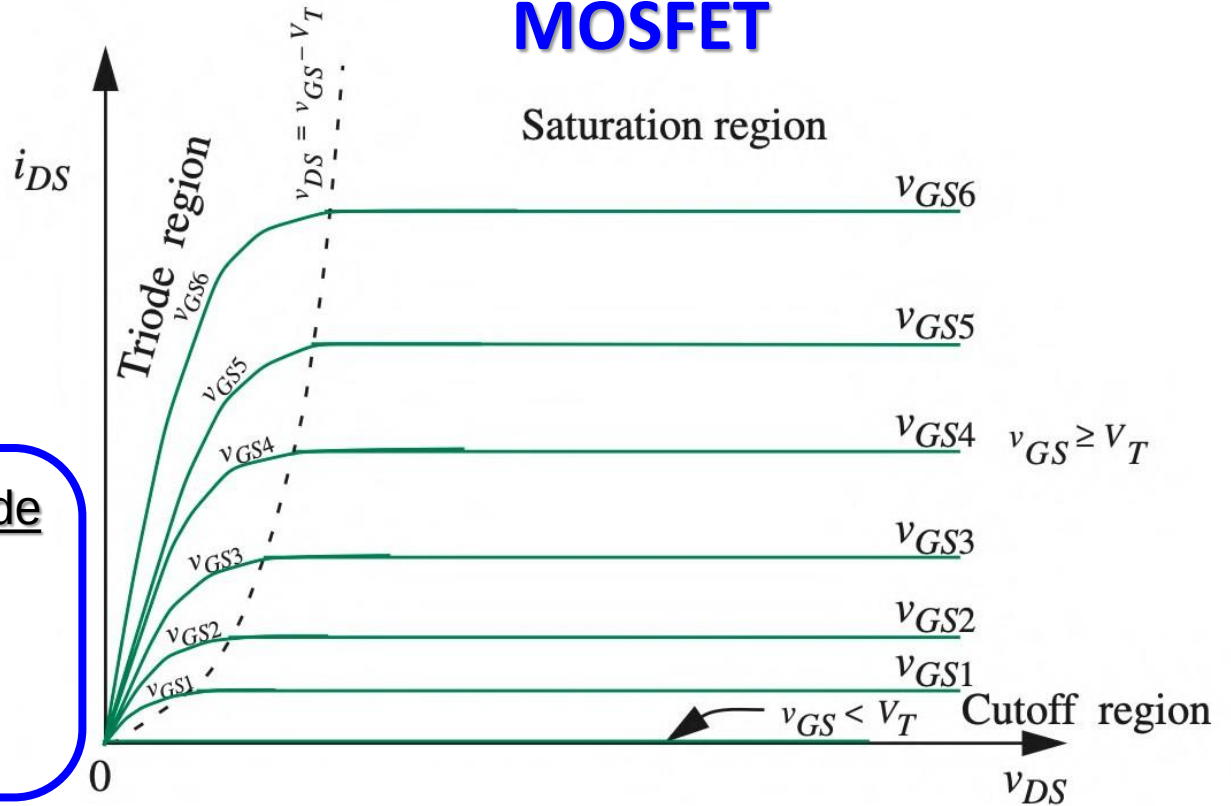
BJT



Slope in the saturation region is **HIGH**, hence very small resistance when "ON"

Slope in the Triode region is **LOW**, hence significant resistance when "ON"

MOSFET

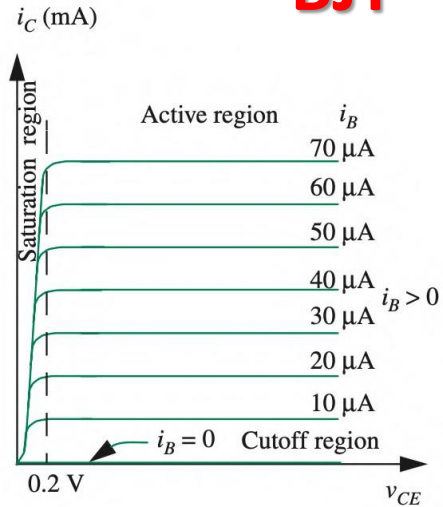


Current in **Saturation** region changes quadratically with control V_{GS} .
Hence, $I_{DS} \propto V_{GS}^2$

Current in **active** region changes linearly with control I_B .
Hence, $I_C \propto I_B$

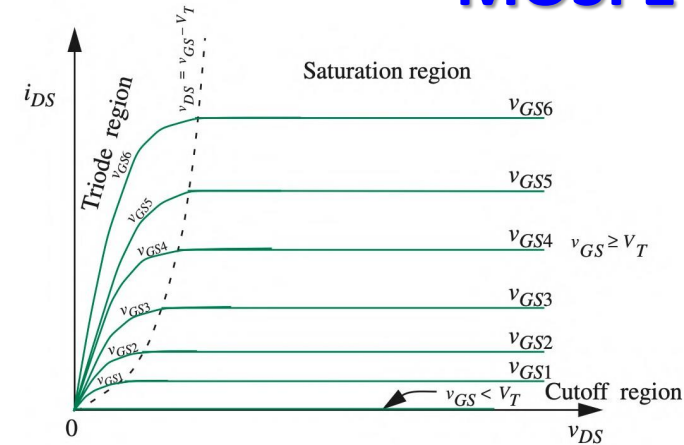
BJT vs MOSFET - Similarities

BJT



- **Saturation** mode for small V_{CE} (< 0.2 V)
- Approximately Short circuit in **Saturation** mode (I_B **HIGH**)
- Open circuit in **Cutoff** mode ($I_B = 0$)
- Can use as a switch $\Rightarrow$ S-Model!

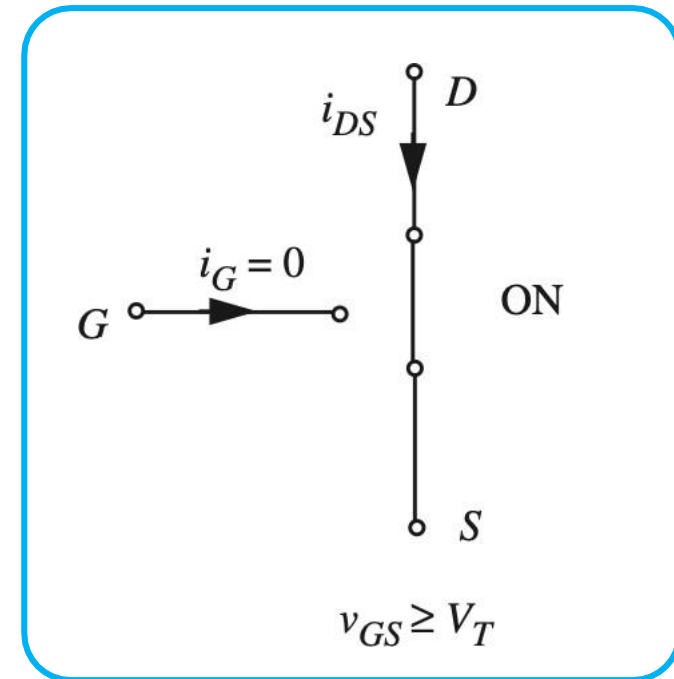
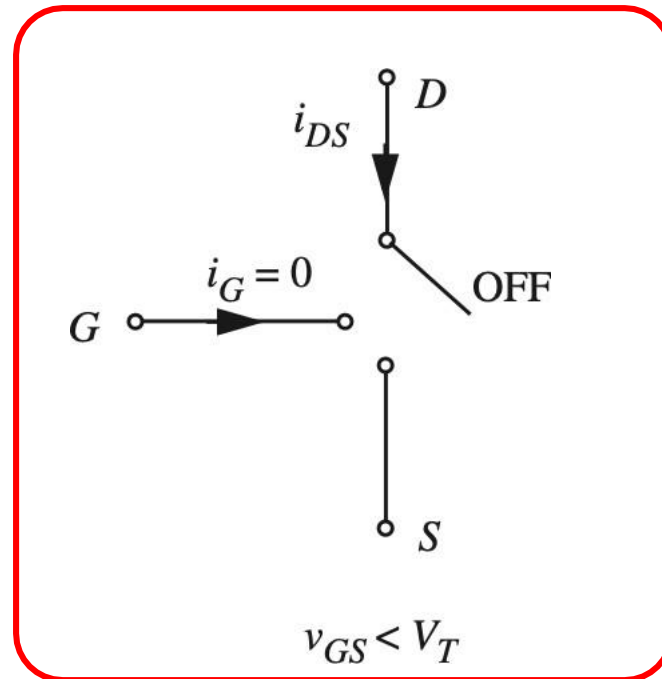
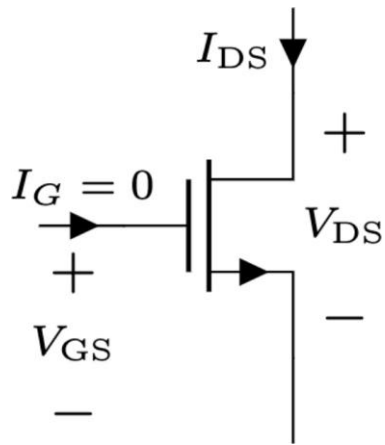
MOSFET



- **Triode** mode for small V_{DS} ($< V_{OV}$)
- Approximately Short circuit in **Triode** mode (V_{GS} **HIGH**)
- Open circuit in **Cutoff** mode ($V_{GS} < V_T = 0$)
- Can use as a switch $\Rightarrow$ S-Model!

MOSFET S-Model

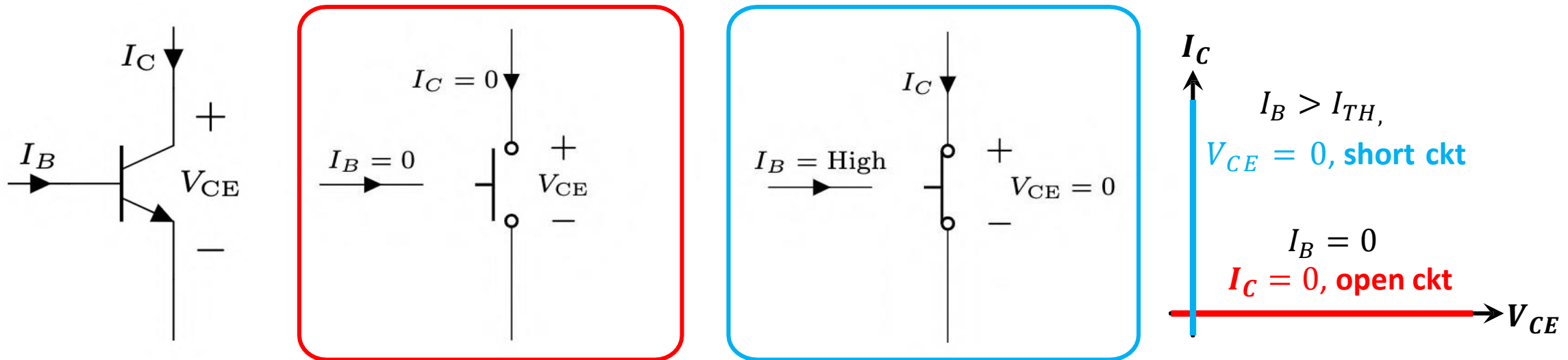
- The MOSFET (approximately) behaves like a switch
- $C = V_{GS}$. Here, $C = \text{"0"} \Rightarrow V_{GS} < V_T$, and $C = \text{"1"} \Rightarrow V_{GS} > V_T$



BJT S-Model

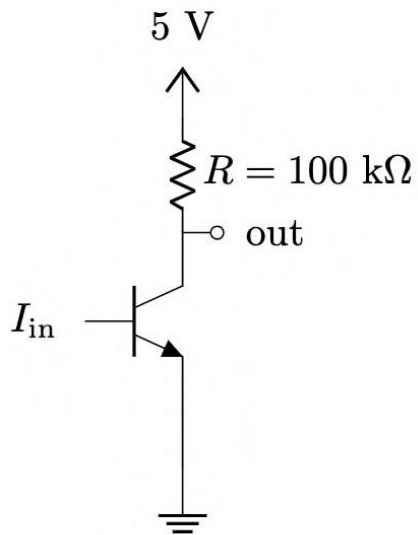
	Representation
Logic 0	$I_B = 0$
Logic 1	$I_B > I_{TH}, I_B = \text{HIGH}$

- The BJT (approximately) behaves like a switch
- $C = I_B$. Here, $C = \text{"0"} \Rightarrow I_B = 0$, and $C = \text{"1"} \Rightarrow I_B > I_{TH}$

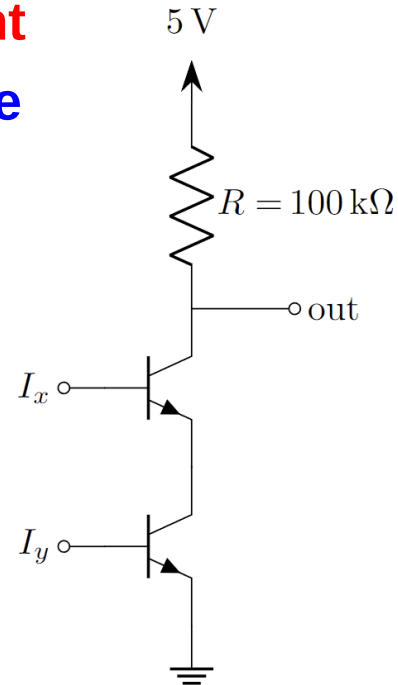


Current-Controlled Logic Gates using BJT

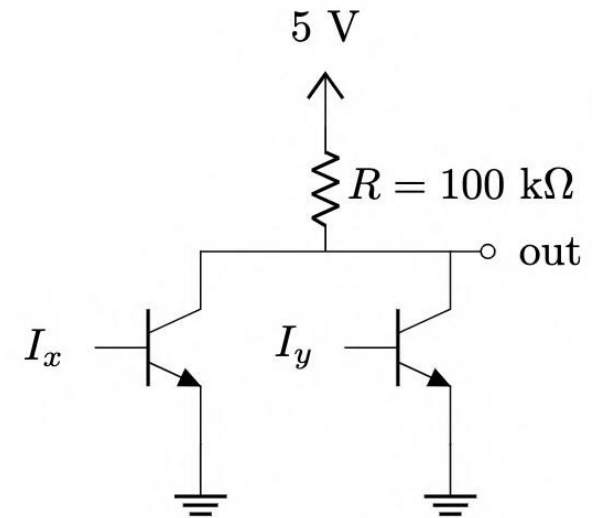
- Just replace switches with BJTs!
- Major problem: Cannot cascade! (Why?)
 - Input Logic Variable: **Current**
 - Output Logic Variable: **Voltage**



BJT Inverter (NOT Gate)



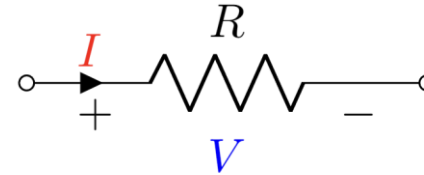
BJT NAND Gate



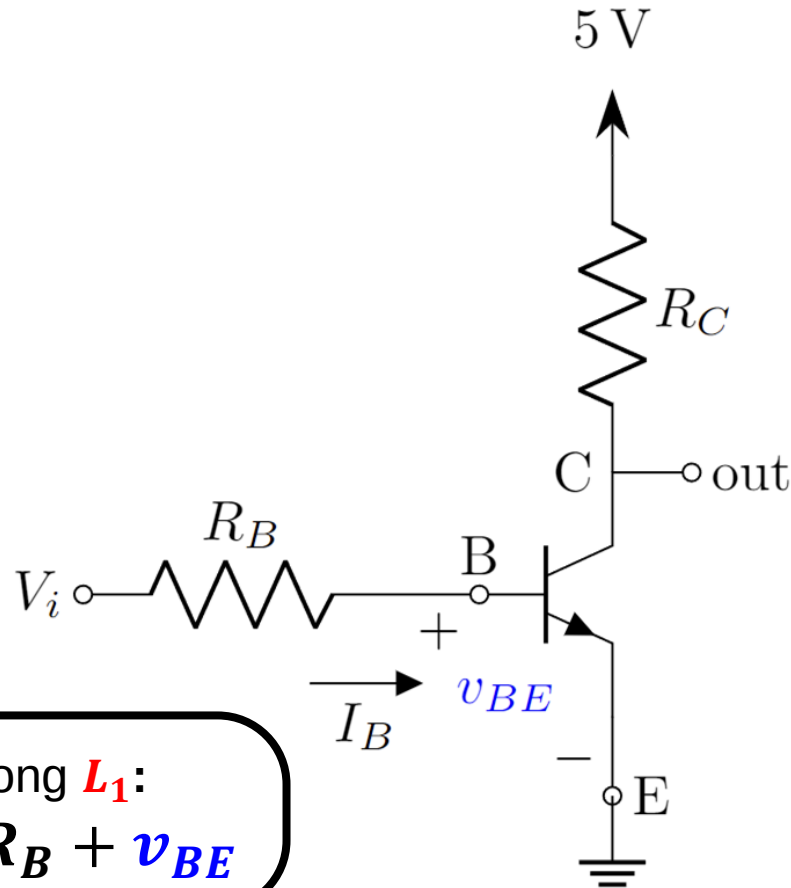
BJT NOR Gate

From Current Controlled to Voltage Controlled

How to convert current I into voltage V ?



$$V = IR$$



$$V_i = I_B R_B + v_{BE}$$

$$I_B = \frac{V_i - v_{BE}}{R_B}$$

v_{BE} depends on I_B .

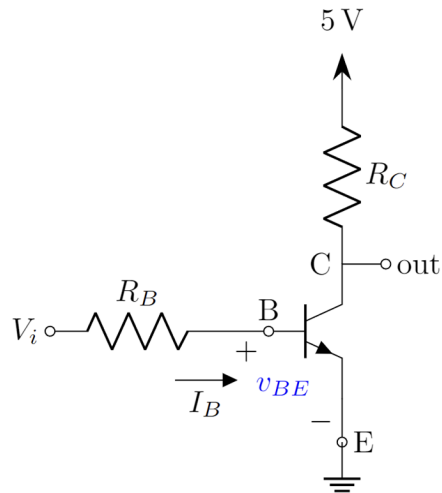
How?

KVL along L_1 :

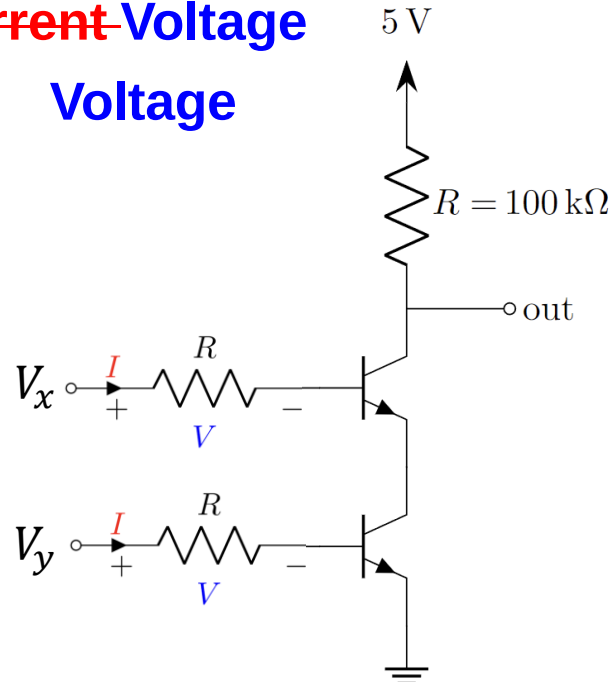
$$V_i = I_B R_B + v_{BE}$$

Logic Gates using BJT

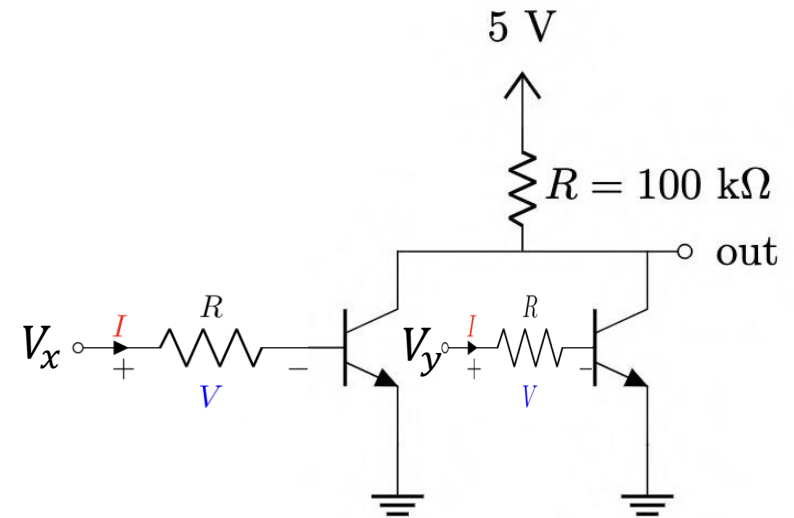
- Just replace switches with BJTs! – and add a Resistor to the Base terminal
- ~~Major problem: Cannot cascade! (Why?)~~ Can be cascaded.
 - Input Logic Variable: ~~Current~~ Voltage
 - Output Logic Variable: Voltage



BJT Inverter (NOT Gate)



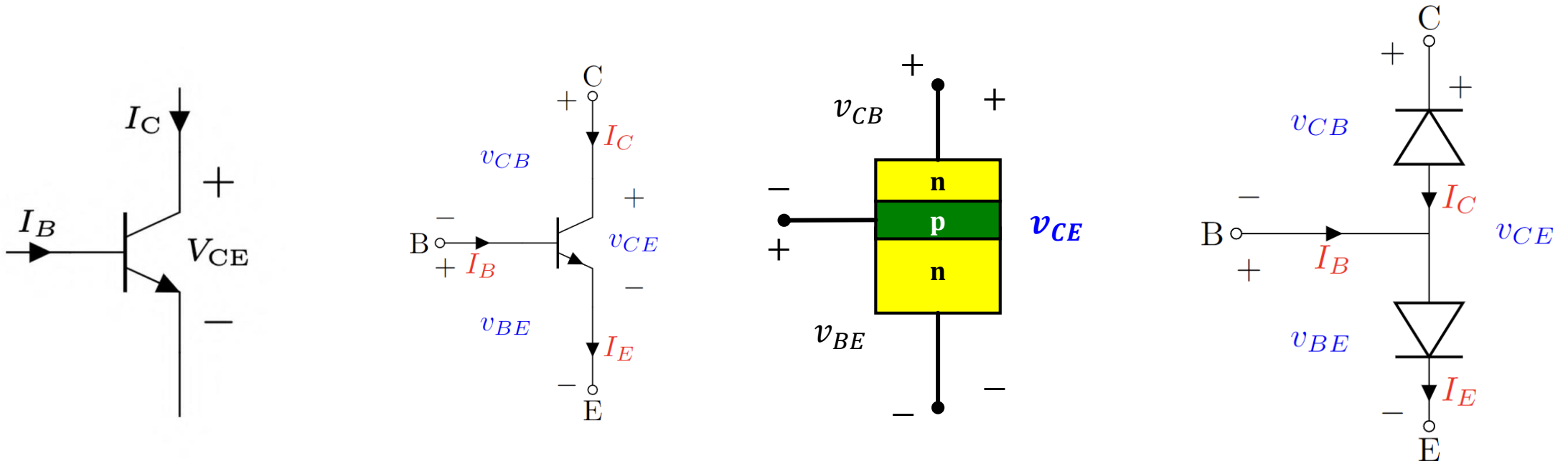
BJT NAND Gate



BJT NOR Gate

Parameters of BJT

A BJT can be thought of as two “pn” junctions placed back-to-back.

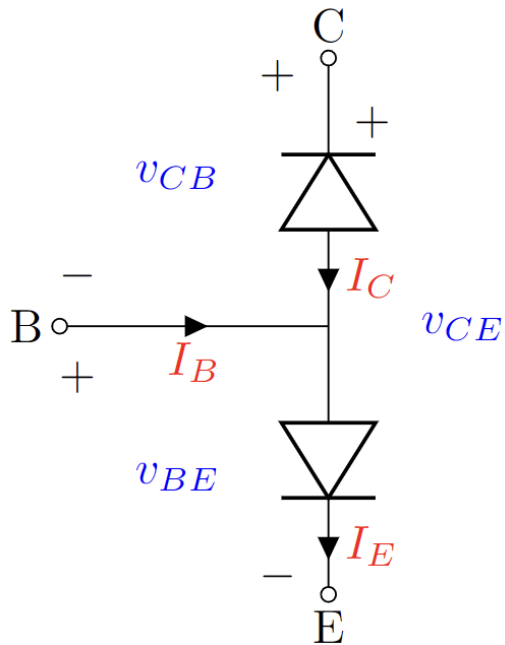


1st pn junction: Across **Base – Emitter:** Voltage v_{BE}

2nd pn junction: Across **Base - Collector:** Voltage v_{BC}

Parameters of BJT

A BJT can be thought of as two “pn” junctions placed back-to-back.



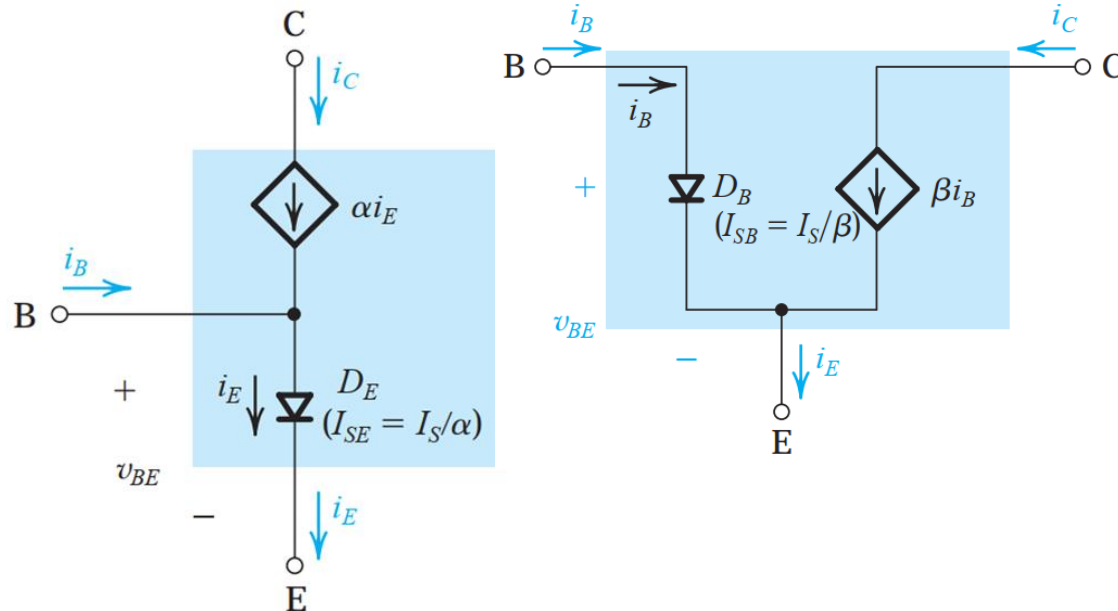
Modes	BE Junction	v_{BE}	CB Junction	v_{CB}	v_{CE}
Cut-off	Reverse Bias	$v_{BE} < 0.7 \text{ V}$	Reverse Bias	$v_{BC} < 0.6 \text{ V}$	
Active	Forward Bias	$v_{BE} = 0.7 \text{ V}$	Reverse Bias	$v_{BC} < 0.6 \text{ V}$	$v_{CE} > 0.2 \text{ V}$
Saturation	Forward Bias	$v_{BE} = 0.8 \text{ V}$	Forward Bias	$v_{BC} = 0.6 \text{ V}$	$v_{CE} = 0.2 \text{ V}$
Reverse Active	Reverse Bias	$v_{BE} < 0.7 \text{ V}$	Forward Bias	$v_{BC} = 0.6 \text{ V}$	$v_{CE} < 0.2 \text{ V}$

1st pn junction: Across **Base – Emitter**: Voltage v_{BE}
 2nd pn junction: Across **Collector - Base**: Voltage v_{CB}

$$v_{CE} = v_{BE} - v_{BC}$$

Parameters of BJT: **Active Mode**

Equivalent circuit of an npn-BJT in **Active Mode**



Current relationships between the three currents in an npn BJT.

$$I_B + I_C = I_E$$

β : Common Emitter Current Gain
 α : Common Base Current Gain

$$I_C = \beta I_B$$

$$I_C = \alpha I_E$$

Only valid for active mode

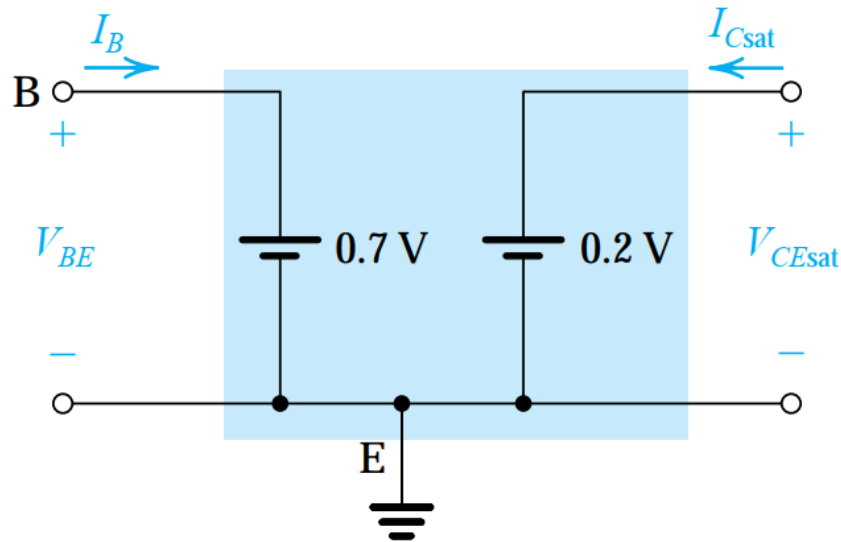
$$\alpha = \frac{\beta}{1 + \beta}$$

1st pn junction: Across **Base – Emitter**: Voltage v_{BE}
2nd pn junction: Across **Base - Collector**: Voltage v_{BC}

$$v_{CE} = v_{BE} - v_{BC}$$

Parameters of BJT: Saturation Mode

Equivalent circuit of an npn-BJT in **Saturation Mode**



Current relationships between the three currents in an npn BJT.

$$I_B + I_C = I_E$$

$$I_{C(sat)} < \beta I_B$$

$$I_C \neq \alpha I_E$$

β : Common Emitter Current Gain
 α : Common Base Current Gain

Only valid for **Saturation** mode

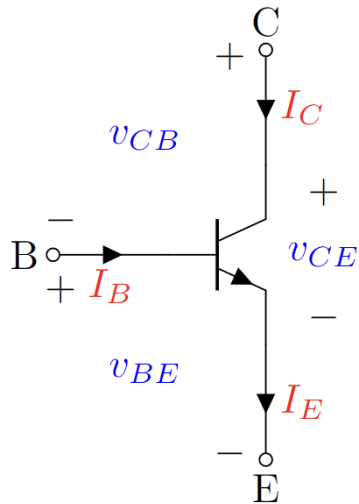
Figure 6.20 A simplified equivalent-circuit model of the saturated transistor.

1st pn junction: Across **Base – Emitter**: Voltage v_{BE}
2nd pn junction: Across **Base - Collector**: Voltage v_{BC}

$$v_{CE} = v_{BE} - v_{BC}$$

Parameters of BJT

A BJT can be thought of as two “pn” junctions placed back-to-back.

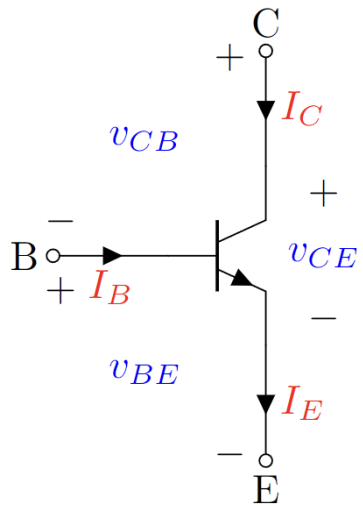


Modes	Conditions!
Cut-off	$v_{BE} < 0.7\text{ V}$ and $v_{BC} < 0.6\text{ V}$
Active	$v_{BE} = 0.7\text{ V}$ and $v_{CE} > 0.2\text{ V}$
Saturation	$v_{BE} = 0.8\text{ V}$ and $v_{CE} = 0.2\text{ V}$ and $\frac{I_C}{I_B} < \beta$
Reverse Active	$v_{BC} = 0.6\text{ V}$ and $v_{EC} > 0.2\text{ V}$

1st pn junction: Across **Base – Emitter**: Voltage v_{BE}
2nd pn junction: Across **Base - Collector**: Voltage v_{BC}

$$v_{CE} = v_{BE} - v_{BC}$$

Parameters of BJT



Modes	Conditions	Equations
Cut-off	$V_{BE} < 0.7 \text{ V} , V_{BC} < 0.6 \text{ V}$	$I_C = I_B = I_E = 0$
Active	$V_{CE} > 0.2 \text{ V}$	$V_{BE} = 0.7 \text{ V}$ $I_C = \beta I_B$ $I_E = (1 + \beta) I_B$ $I_C = \alpha I_E$
Saturation	$\frac{I_C}{I_B} < \beta$	$V_{BE} = 0.8 \text{ V}$ $V_{CE} = 0.2 \text{ V}$

1st pn junction: Across **Base – Emitter**: Voltage v_{BE}
 2nd pn junction: Across **Base - Collector**: Voltage v_{BC}

Solving Circuits with **BJT**

- Use **Method of Assumed State!**
- **Three steps:**
 - **Assume:** One of the modes
(Cutoff, **Saturation**, **Active**)
 - **Solve:** Use corresponding equation and KCL + KVL with currents
 - **Verify:** Check if the conditions of v_{BE} and v_{CE} are satisfied. If not, repeat.
- Might need to solve quadratic equation ($ax^2 + bx + c = 0$).
- If we get two roots, choose the one that's *favorable* to your assumption

BJT Problem 1

Analyze the circuit to find I_C and v_{out} using the Method of Assumed State. Here, the input of the BJT is $V_i = 1\text{ V}$. You must validate your assumptions.

Assume:

Let the BJT be in **ACTIVE** mode

So, $v_{BE} = 0.7\text{ V}$
 $v_{CE} > 0.2\text{ V}$

Solve:

Equations: $I_B = \frac{V_i - v_{BE}}{R_B} = \frac{1 - 0.7}{100} \text{ mA} = 3\text{ }\mu\text{A}$

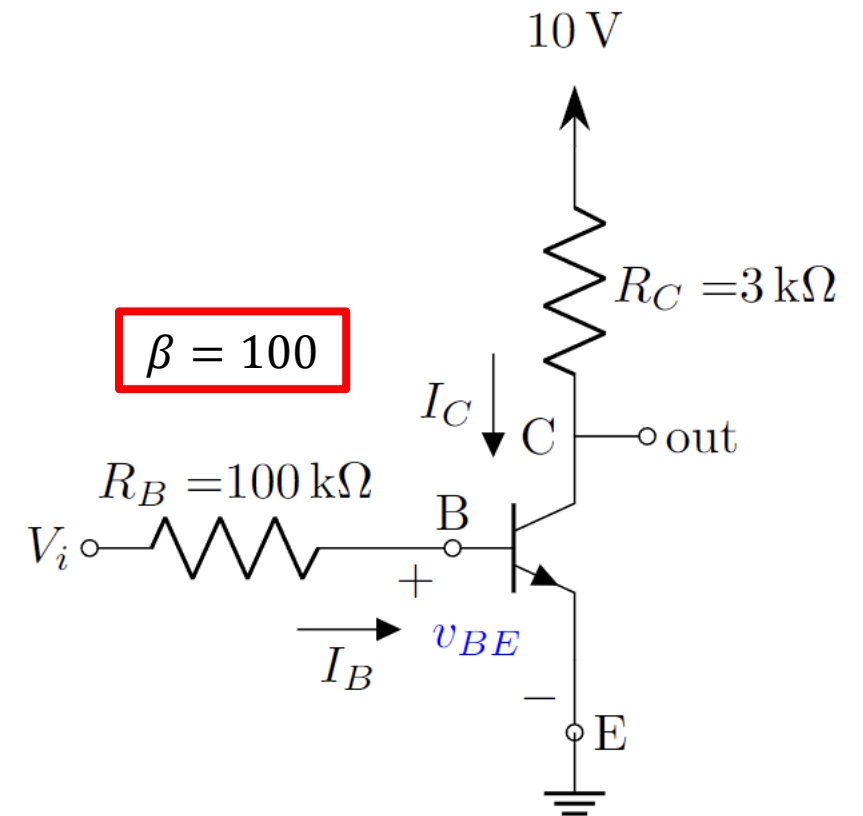
$$I_C = \beta I_B = 100 \times 3 \times 10^{-3} \text{ mA} = 0.3\text{ mA}$$

$$v_{out} = 10 - I_C R_C = (10 - 0.3 \times 3) \text{ V} = 9.1\text{ V}$$

Verify: For **ACTIVE** condition $\rightarrow v_{CE} > 0.2\text{ V}$

Here, $v_{CE} = v_{out} = 9.1\text{ V} > 0.2\text{ V}$

Assumption is Correct!



BJT Problem 2

Analyze the circuit to find I_C and v_{out} using the Method of Assumed State. Here, the input of the BJT is $V_i = 5\text{ V}$. You must validate your assumptions.

Assume:

Let the BJT be in **ACTIVE** mode

So, $v_{BE} = 0.7\text{ V}$
 $v_{CE} > 0.2\text{ V}$

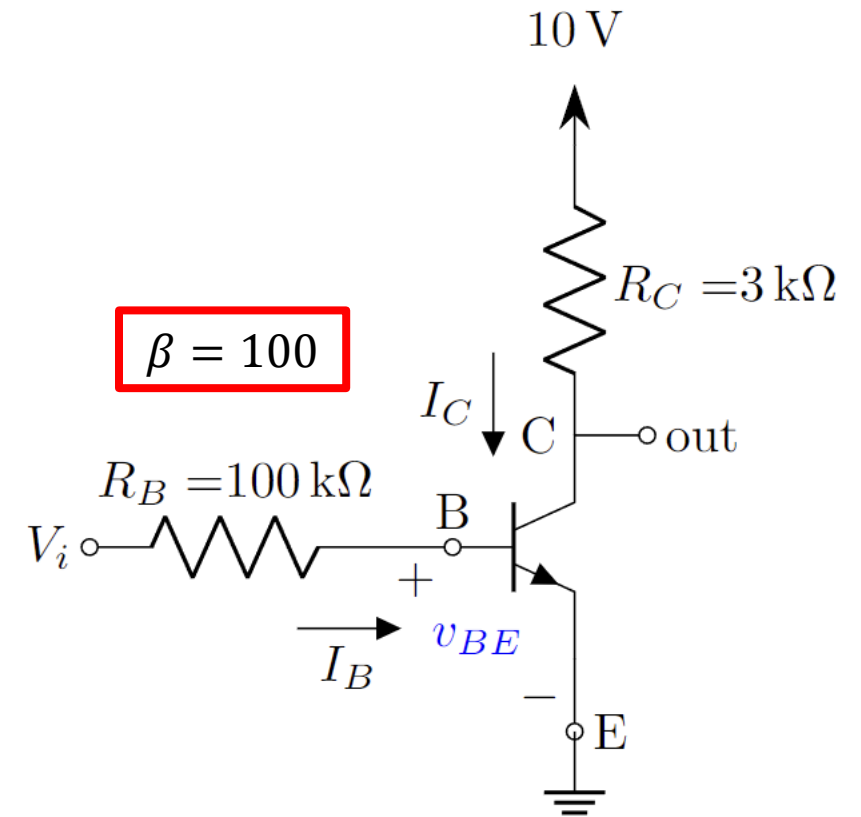
Solve:

Equations:

$$I_B = \frac{V_i - v_{BE}}{R_B} = \frac{5 - 0.7}{100} \text{ mA} = 43 \mu\text{A}$$
$$I_C = \beta I_B = 100 \times 43 \times 10^{-3} \text{ mA} = 4.3 \text{ mA}$$
$$v_{out} = 10 - I_C R_C = (10 - 4.3 \times 3) \text{ V} = -2.9 \text{ V}$$

Verify: For **ACTIVE** condition $\rightarrow v_{CE} > 0.2\text{ V}$
Here, $v_{CE} = v_{out} = -2.9\text{ V} \not> 0.2\text{ V}$

Assumption is Wrong!



BJT Problem 2

Analyze the circuit to find I_C and v_{out} using the Method of Assumed State. Here, the input of the BJT is $V_i = 5\text{ V}$. You must validate your assumptions.

Assume:

Let the BJT be in **Saturation** mode

So, $v_{BE} = 0.8\text{ V}$ and $\frac{I_C}{I_B} < \beta$

$$v_{CE} = 0.2\text{ V}$$

Solve:

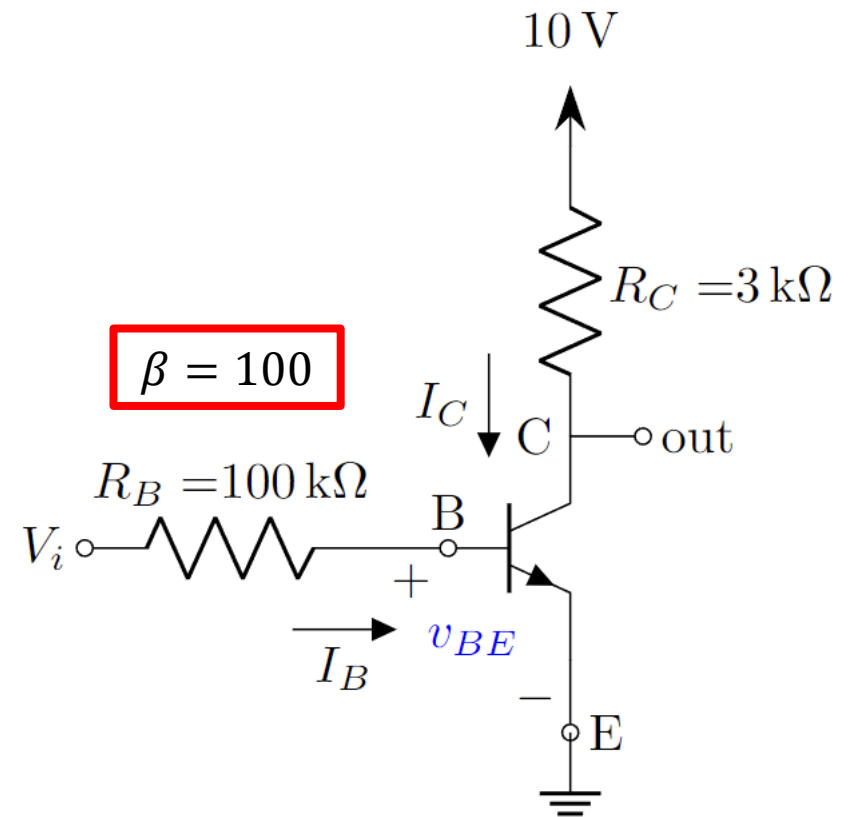
Equations:
$$I_B = \frac{V_i - v_{BE}}{R_B} = \frac{5 - 0.8}{100} \text{ mA} = 43\text{ }\mu\text{A}$$

$$I_C = \beta I_B \frac{10 - v_{CE}}{R_C} = \frac{10 - 0.2}{3} \text{ mA} = 3.27\text{ mA}$$

$$v_{out} = v_{CE} = 0.2\text{ V}$$

Verify: For **Saturation** condition $\rightarrow \frac{I_C}{I_B} < \beta$

Here, $\beta = 100$
$$\frac{I_C}{I_B} = \frac{3.27}{0.043} = 76 < 100$$



Assumption is Correct!

BJT Problem 3

Analyze the circuit to find I_B , I_C , I_E and v_o using the Method of Assumed State. You must validate your assumptions.

Assume:

Let the BJT be in **ACTIVE** mode

So,

$$v_{BE} = 0.7 \text{ V}$$
$$v_{CE} > 0.2 \text{ V}$$

Solve:

Equations:

$$V_i = I_B R_B + V_{BE} + I_E R_E = I_B R_B + V_{BE} + (1 + \beta) I_B R_E$$

$$I_B = \frac{V_i - V_{BE}}{R_B + (1 + \beta) R_E} = \frac{5 - 0.7}{10 + (1 + 100) 2} = 0.0203 \text{ mA}$$

$$I_C = \beta I_B = 2.03 \text{ mA}$$

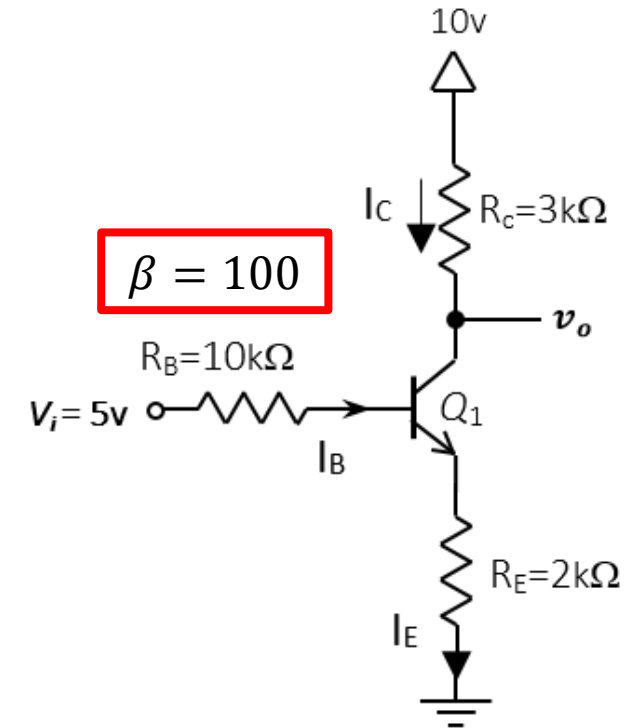
$$I_E = I_B + I_C = 2.05 \text{ mA}$$

$$10 = I_C R_C + V_{CE} + I_E R_E$$

$$V_{CE} = -1.9$$

Verify: For **ACTIVE** condition $\rightarrow V_{CE} > 0.2 \text{ V}$

Here $V_{CE} < 0.2 \text{ V}$ **Assumption is Wrong!**



BJT Problem 3

Analyze the circuit to find I_B , I_C , I_E and v_o using the Method of Assumed State. You must validate your assumptions.

Assume:

Let the BJT be in **Saturation** mode

So, $v_{BE} = 0.8 \text{ V}$ and $\frac{I_C}{I_B} < \beta$
 $v_{CE} = 0.2 \text{ V}$

Solve:

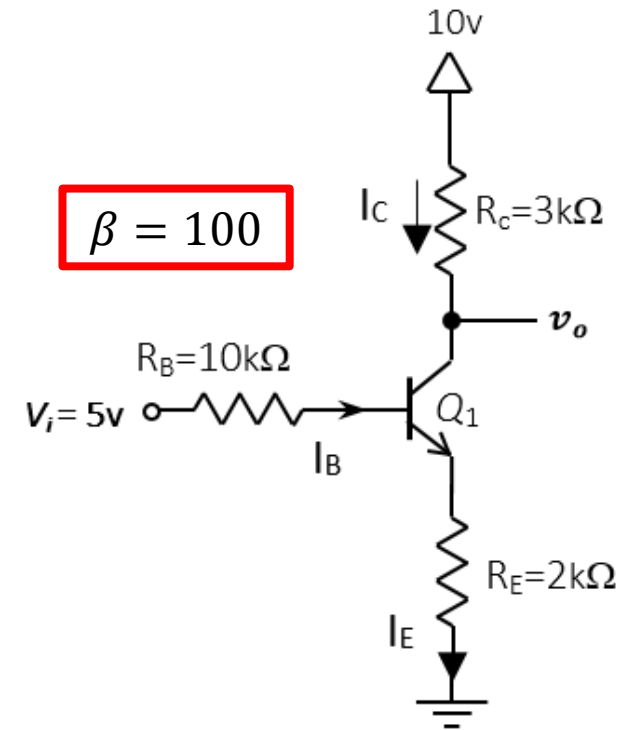
Equations:

$$5 = 10I_B + 0.8 + 2I_E$$
$$10I_B + 0.1I_C + 2I_E = 4.2 \quad \dots \dots \dots (a)$$
$$10 = 3I_C + 0.2 + 2I_E$$
$$0.1I_B + 3I_C + 2I_E = 9.8 \quad \dots \dots \dots (b)$$
$$I_B + I_C - I_E = 0 \quad \dots \dots \dots (c)$$

Solving (a), (b) & (c) $I_B = 0.025 \text{ mA}$ $I_C = 1.95 \text{ mA}$ $I_E = 1.975 \text{ mA}$

Verify: For **Saturation** condition $\rightarrow \frac{I_C}{I_B} < \beta$

$$\frac{I_C}{I_B} = \frac{1.95}{0.025} = 78 < 100$$
$$v_o = 10 - I_C R_C = 4.15 \text{ V}$$



Assumption is Correct!

BJT Problem 4

Analyze the circuit to find I_B , I_C , I_E and v_{out} using the Method of Assumed State. You must validate your assumptions.

Assume:

Let the BJT be in **Active** mode

So,

$$v_{BE} = 0.7 \text{ V}$$

$$v_{CE} > 0.2 \text{ V}$$

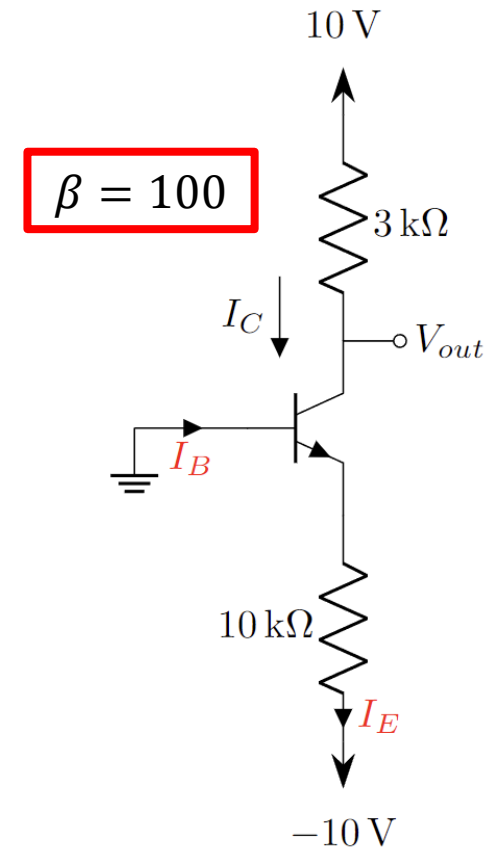
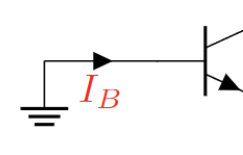
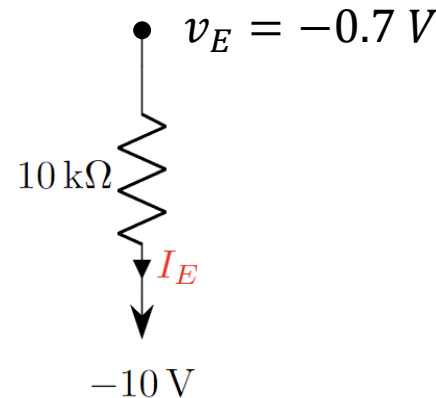
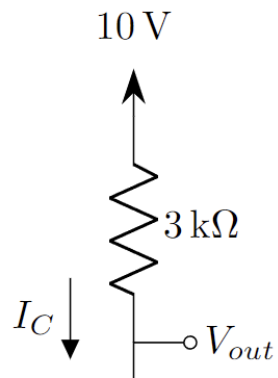
Solve:

Equations:

$$I_E = \frac{v_E - (-10)}{10} = \frac{-0.7 + 10}{10} \text{ mA} = \mathbf{0.93 \text{ mA}}$$

$$I_B = \frac{1}{\beta} I_C = \frac{1}{\beta} \cdot \alpha I_E = \frac{1}{\beta} \cdot \frac{\beta}{\beta + 1} I_E = \frac{1}{\beta + 1} I_E = \mathbf{9.21 \text{ }\mu\text{A}}$$

$$\begin{aligned} v_{out} = v_C &= 10 - 3I_C = 10 - 3\beta I_B \\ &= (10 - 3 \cdot 100 \cdot 9.207 \times 10^{-3}) \text{ V} \\ &= \mathbf{7.237 \text{ V}} \end{aligned}$$



BJT Problem 4

Analyze the circuit to find I_B , I_C , I_E and v_{out} using the Method of Assumed State. You must validate your assumptions.

Assume:

Let the BJT be in **Active** mode

So, $v_{BE} = 0.7 \text{ V}$
 $v_{CE} > 0.2 \text{ V}$

Solve:

Equations:

$$I_E = 0.93 \text{ mA}$$

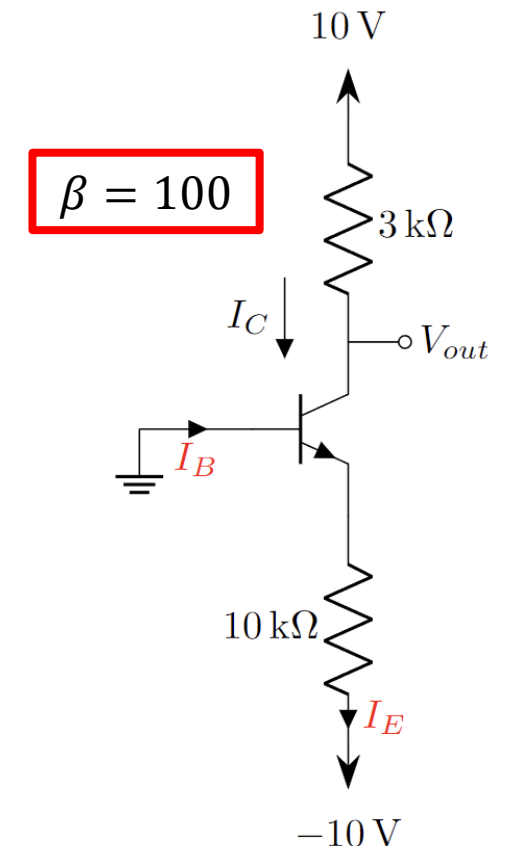
$$I_B = 9.21 \text{ }\mu\text{A}$$

$$v_{out} = v_C = 7.237 \text{ V} \quad v_B = 0 \text{ V} \quad v_E = -0.7 \text{ V}$$

Verify: For **ACTIVE** condition $\rightarrow v_{CE} > 0.2 \text{ V}$

Here, $v_{CE} = (7.237 + 0.7) \text{ V} = 7.937 \text{ V} > 0.2 \text{ V}$

Assumption is Correct!



BJT Problem 5

Analyze the circuit to find I_B , I_C , I_E and v_o using the Method of Assumed State. You must validate your assumptions.

Assume:

Let the BJT be in **Active** mode

So,

$$v_{BE} = 0.7 \text{ V}$$
$$v_{CE} > 0.2 \text{ V}$$

Solve:

Equations:

$$V_i = I_B R_B + V_{BE} + I_E R_E - 1.8 = I_B R_B + V_{BE} + (1 + \beta) I_B R_E - 1.8$$

$$I_B = \frac{V_i - V_{BE} + 1.8}{R_B + (1 + \beta) R_E} = \frac{1 - 0.7 + 1.8}{560 + (1 + 75) 3} = 2.66 \mu\text{A}$$

$$I_C = \beta I_B = 0.2 \text{ mA}$$

$$I_E = I_B + I_C = 0.203 \text{ mA}$$

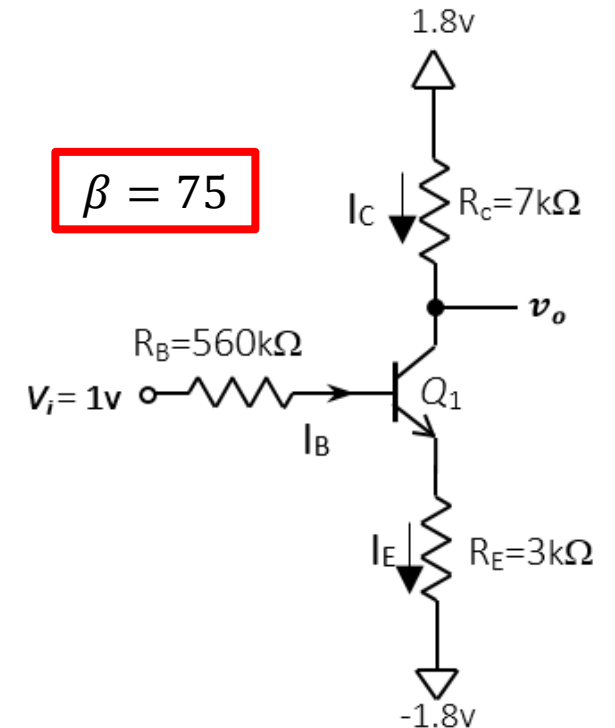
$$1.8 = I_C R_C + V_{CE} + I_E R_E - 1.8$$

$$V_{CE} = 1.59$$

Verify: For **ACTIVE** condition $\rightarrow V_{CE} > 0.2 \text{ V}$

Here $V_{CE} > 0.2 \text{ V}$

Assumption is Correct!



BJT Problem 6

Analyze the circuit to determine the voltages at all nodes and current through all branches. Assume $\beta = 100$

Assume:

Let the BJT be in **Active** mode

So, $v_{BE} = 0.7 \text{ V}$
 $v_{CE} > 0.2 \text{ V}$

Solve:

Equations:

$$15 = 100I + 0.7 + 3I_E$$

$$15 = 100I + 50I_1$$

$$I = I_B + I_1$$

$$I_E = (1 + \beta)I_B$$

$$100I_1 + 403I_B = 14.3 \dots\dots\dots (a)$$

$$150I_1 + 100I_B = 15 \dots\dots\dots (b)$$

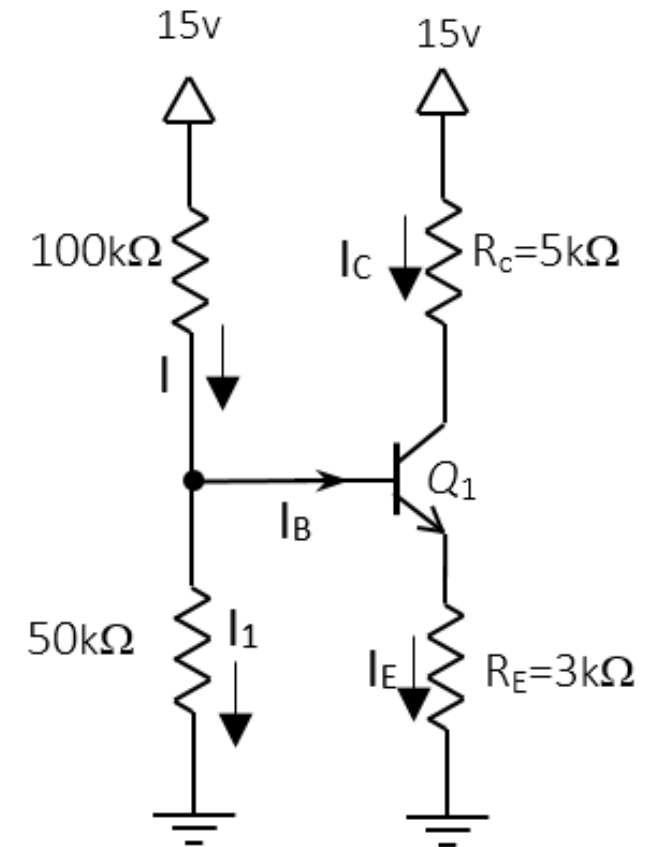
Solving (a) and (b)

$$I_B = 0.013 \text{ mA}$$

$$I_C = \beta I_B = 1.3 \text{ mA}, I_E = 1.313 \text{ mA}$$

$$15 = I_C R_C + V_{CE} + I_E R_E$$

$$V_{CE} = 4.561$$



BJT Problem 6

Analyze the circuit to determine the voltages at all nodes and current through all branches. Assume $\beta = 100$

Verify: For **ACTIVE** condition $\rightarrow v_{CE} > 0.2 \text{ V}$

Here, $v_{CE} = 4.561 \text{ V} > 0.2 \text{ V}$

Assumption is Correct!

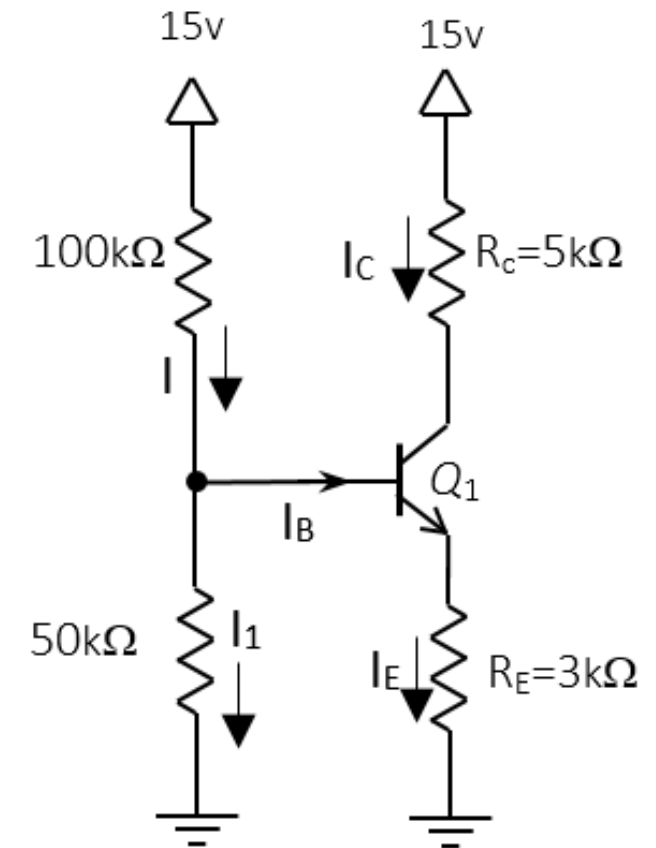
$$v_B = 0.7 + 3I_E = 4.639\text{v}$$

$$I_C = \frac{15 - v_c}{5}$$

$$v_c = 8.5\text{v}$$

$$I_E = \frac{v_E - 0}{3}$$

$$v_E = 3.94\text{v}$$



BJT Problem 6

Analyze the circuit to determine the voltages at all nodes and current through all branches. Assume $\beta = 100$

Alternative Method

Simplifying the base circuit using **Thevenin's Theorem**

